

Fatal Pedestrian & Cyclist Crash Risk Factors in Oregon

A Spatial Analysis (2015-2024)

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Abstract

Using 2015-2024 FARS records enriched with Census data, we identified where fatal pedestrian and cyclist crashes cluster across Oregon and examined the roadway and neighborhood characteristics associated with them. The results highlight recurring high-risk corridors and socioeconomic patterns that can help transportation agencies prioritize further investigation and targeted safety investment.



Introduction & Motivation

Problem & question

Pedestrians and cyclists are physically vulnerable in roadway collisions, and fatal crashes may reflect recurring patterns in road design, lighting, speed, location, and surrounding community conditions. We ask which environmental, roadway, and temporal factors are most strongly associated with fatal pedestrian and cyclist crashes in Oregon from 2015-2024, and whether these crashes are disproportionately concentrated in communities with particular demographic or socioeconomic characteristics.

Impact & stakeholders

The findings may support ODOT, city and county planners, public-health organizations, and community advocates in identifying locations and conditions that warrant closer safety review. Pedestrians and cyclists are the primary affected stakeholders, while transportation agencies are positioned to act on the results.



Background & Related Work

FARS provides standardized national records for police-reported motor-vehicle crashes involving at least one fatality [1]. Transportation-safety studies commonly combine crash records with spatial hotspot analysis, roadway characteristics, and statistical modeling to identify high-risk locations and contributing conditions.

This project adds neighborhood context by spatially joining fatal crash locations to Census tracts and ACS indicators [2]. It differs from a purely descriptive crash map by combining spatial clustering, statistical testing, and an equity-focused analysis within one statewide Oregon dataset.



Methods & Evaluation

Approach

Spatial Hotspot & Infrastructure Analysis: We applied DBSCAN with a 200 meter radius and a minimum of two fatalities to identify localized crash hotspots while filtering isolated incidents. Each cluster was then profiled using its most common roadway characteristics – including road type, intersection status, trafficway design, speed limit, and lighting – to identify recurring infrastructure patterns.

Demographic Correlation: We calculated Pearson correlation coefficients to measure the linear relationship between Census tract demographics (e.g., median income, % Black population) and the tract-level fatal crash rate.

Validation

Spatial Validation: We cross-referenced our mathematically derived hotspots against satellite imagery and ODOT crash logs to confirm the presence of high-risk infrastructure (e.g., multi-lane arterials, mid-block crossings).

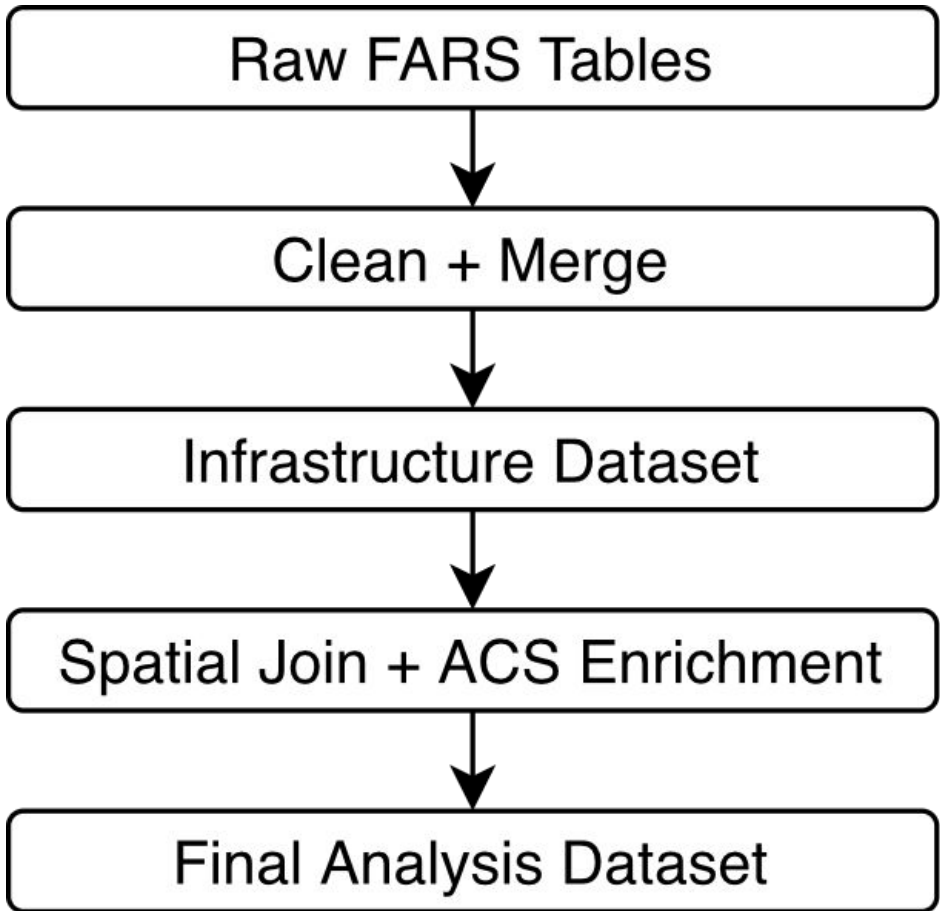


Data, Engineering & Ethics

Sources & scale

The final dataset contains **1,049 fatal pedestrian and cyclist victim records**, representing **490 unique Oregon incidents** from 2015–2024. It combines NHTSA FARS accident, person, pedestrian/bicyclist, vehicle, and weather tables with Census tract boundaries and ACS demographic indicators. The final analysis-ready dataset contains 38 variables.

Pipeline



Ethics & responsible use

The project uses public, anonymized government data with no direct personal identifiers. FARS includes only crashes involving at least one fatality, so findings describe fatal-crash patterns rather than total pedestrian or cyclist risk. Census variables describe neighborhoods, not individual victims, and associations should not be interpreted as causation. Results should guide equitable investigation and investment rather than stigmatize communities identified as hotspots.



Discussion, Limitations & Conclusions

Discussion

Eleven hotspots revealed recurring arterial and non-intersection conditions. Crash rates were also significantly associated with income, Black population share, and transit or walking commute share; these relationships are associative, not causal.

Limitations & future work

FARS excludes non-fatal crashes and lacks exposure measures. Missing roadway data and small cluster sizes limit interpretation. Future work should add non-fatal incidents, traffic and active-travel volumes, and corridor-level infrastructure data.

Conclusion

Fatal pedestrian and cyclist crashes in Oregon concentrate in 11 localized hotspots and frequently occur on arterial roads away from intersections. Measurable neighborhood-level disparities further suggest that agencies should prioritize both high-risk corridors and potentially underserved communities for detailed safety review and targeted investment.



Results & Visualizations

Infrastructure Analysis

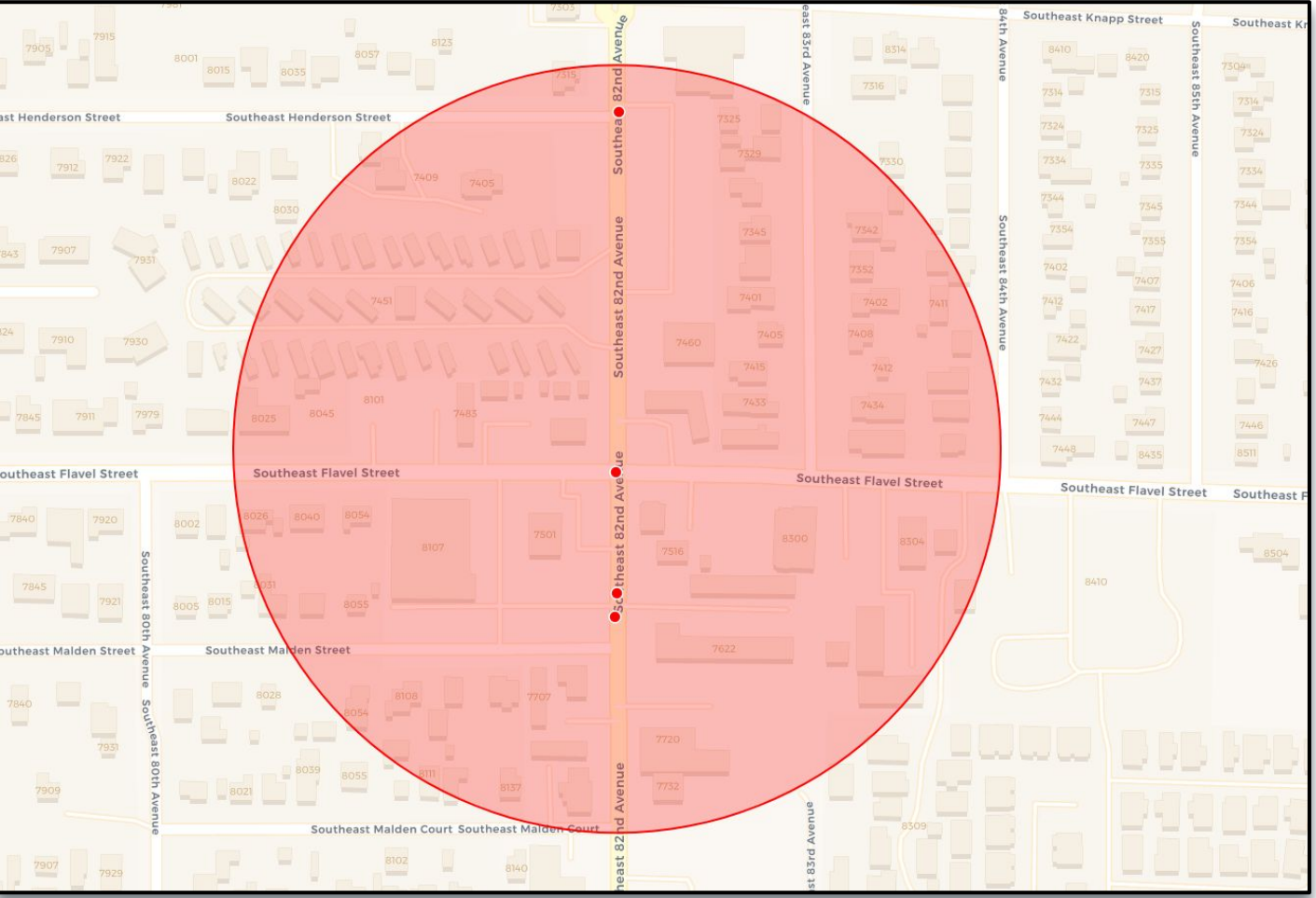


Fig. 1: Cluster #1 of 11

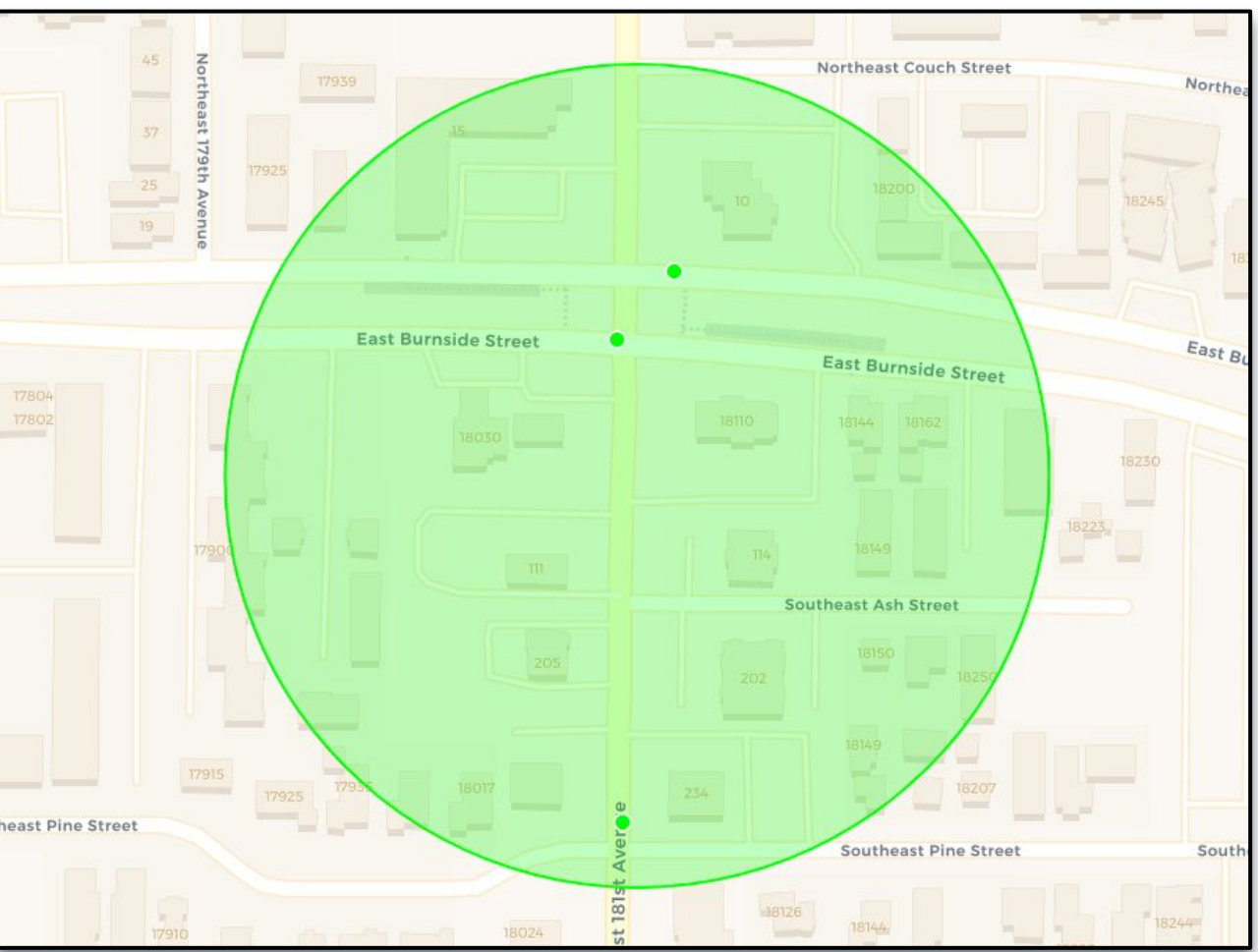


Fig. 2: Cluster #2 of 11

“Reactivated Hotspot”

Southeast 82nd Avenue, Portland, OR

Unique Incidents: 4

Total Fatalities: 4

Radius: 163.1m

Years Active:
2016, 2017, 2018, 2024

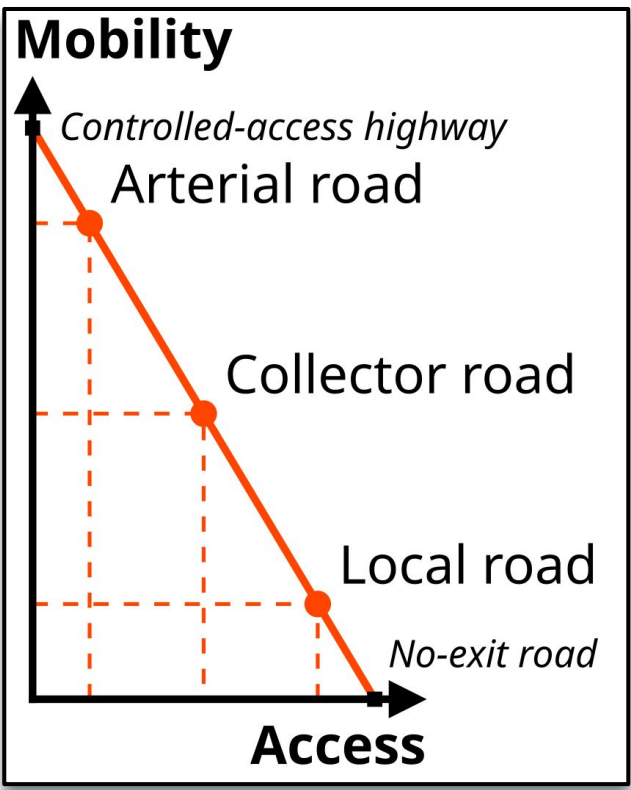


Fig. 3: Road Hierarchy Mobility

“Cyclical Hotspot”

Southeast 181st Avenue, Portland, OR

Unique Incidents: 3

Total Fatalities: 3

Radius: 123.6m

Years Active:
2015, 2019, 2023

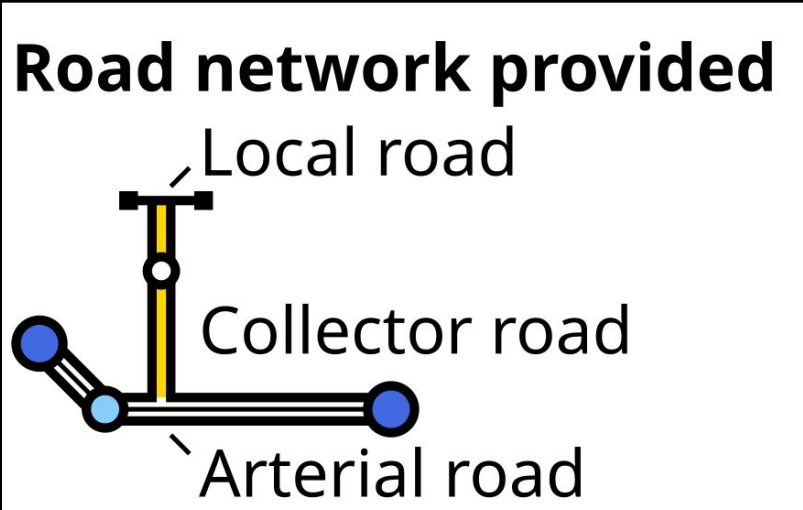


Fig. 4: Road Hierarchy Connection

Feature	Percentage Occurrence
Arterial Road	92.5%
Not an Intersection	57.5%
Two-Way With a Continuous Left-Turn Lane	50.0%

Fig. 5: Cluster Feature Analysis

Feature	Percentage Occurrence
Arterial Road	60.4%
Not an Intersection	70.6%
Two-Way With a Continuous Left-Turn Lane	23.7%

Fig. 6: Full Statewide Feature Analysis

Demographic Analysis

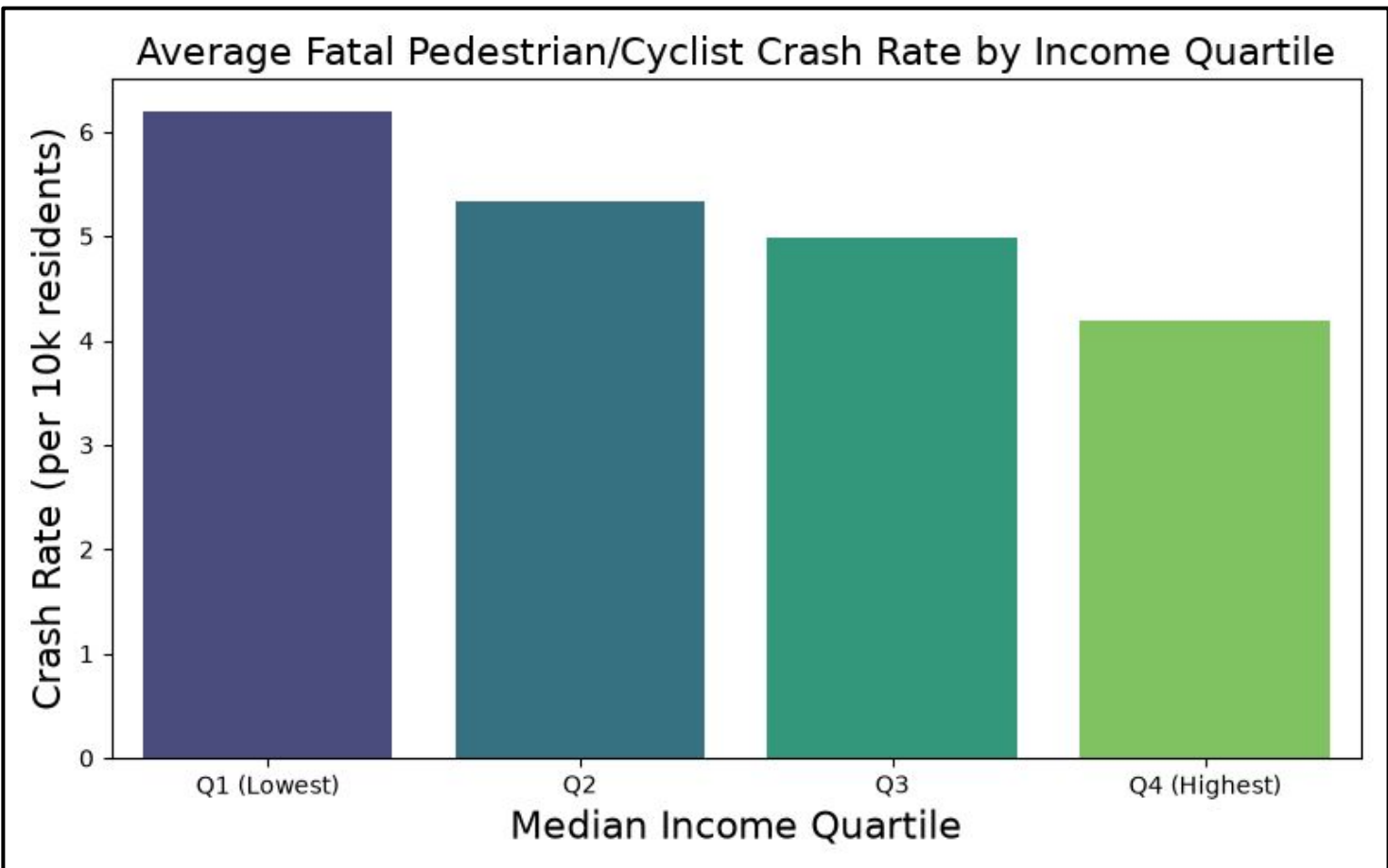


Fig. 7: Average Fatal Crash Rate by Income

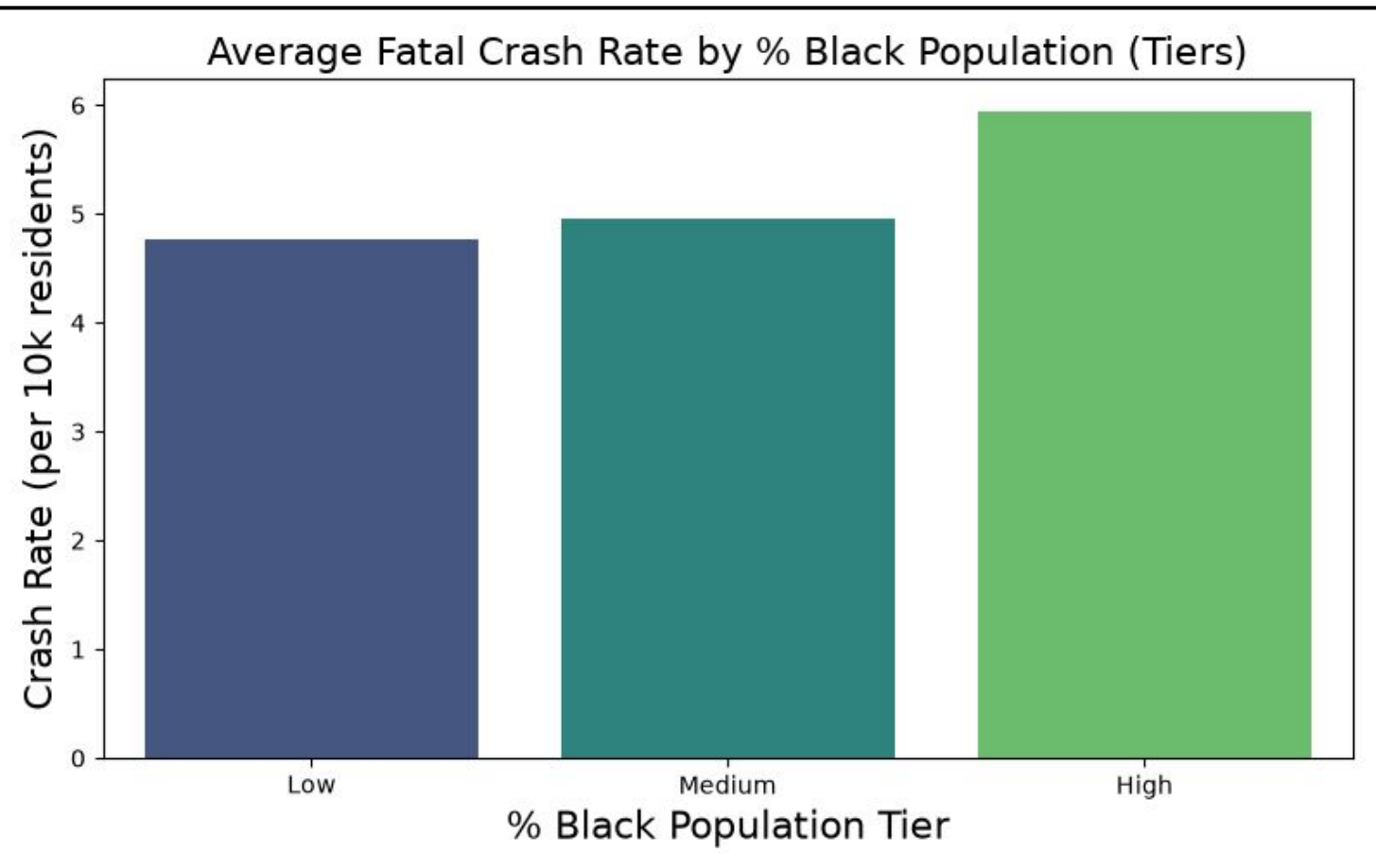


Fig. 8: Average Fatal Crash Rate by Black Population

Variable	Correlation (r)	Significance (p < 0.05)	Interpretation
PCT_BLACK	0.184	<0.001	Weak positive
PCT_TRANSIT_WALK_COMMUTE	0.143	0.002	Weak positive
MEDIAN_INCOME	-0.136	0.003	Weak negative
PCT_WHITE	-0.077	0.090	Not significant, but close
PCT_HISPANIC	0.008	0.868	Not significant

Fig. 9: Correlation with Crash Rate per 10k Population



Reproducibility & References

Repository – <https://github.com/S1mT/data510-OregonPedSafety> · Key dependencies – Python 3.x, pandas, geopandas, folium, shapely, numpy, scikit-learn, seaborn, matplotlib, requests, scipy, os

References: [1] National Highway Traffic Safety Administration, Fatality Analysis Reporting System (FARS), <https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system-fars>, 2026. [2] U.S. Census Bureau, American Community Survey 5-Year Data, <https://www.census.gov/data/developers/data-sets/acs-5year.html>, 2026.

[3] Oregon Department of Transportation, Oregon Bicycle and Pedestrian Plan, <https://www.oregon.gov/ODOT/Planning/Documents/OBPP.pdf>, 2016.

